Certifying Causal Scaling: A Metamorphic Verifier for Learned Physical Predictors

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Abstract

Learned physical predictors enter systems when simulation is too slow to run online, but spending compute to scale them is only justified if it buys causal understanding rather than memorization of the training distribution. Accuracy alone cannot certify that, because a model can fit its training task while violating the physics it claims to model. We build a metamorphic verifier that certifies causal scaling. A battery of five physics-relation tests scores each trained predictor into a Causal Robustness Score, and a scaling detector fits power laws of these errors against compute to emit a verdict. The battery is validated against a perfect causal oracle, the exact solution of the training task, and on a GPU-scale grid of sixteen size-by-budget cells the verifier certifies causal scaling on a bouncing-ball benchmark: causal error falls with slope -0.111 while task accuracy saturates at 0.92, the opposite of the stochastic-parrot signature. The certificate turns "is scaling working?" into a falsifiable, reproducible verdict.

1 Introduction

Learned physical predictors are attractive to systems builders because a trained surrogate can answer a dynamics query faster than the simulator that produced its data (Sanchez-Gonzalez et al., 2020). They are also expensive: a single production-quality predictor is a GPU training run, and the standard question an operator asks before spending that run is whether more parameters and more data will keep paying off. Scaling-law studies gave the field a precise answer for task accuracy, which falls as a power law of compute in language and vision (Kaplan et al., 2020; Hoffmann et al., 2022), but a power law on task error does not say what the added capacity is being used for. A model can fit its training task better by memorizing the training distribution, and the stochastic-parrot critique of large models is exactly this failure mode: fluent output without the underlying competence (Bender et al., 2021). For a physical predictor the competence is the physics itself, so the question of whether scaling buys understanding or memorization is a question about whether the model's outputs transform covariantly with the laws of motion.

The obstruction is measurement. Task accuracy is easy to measure and impossible to interpret causally, because a memorizing model and a causal model can reach the same training error (Zhang et al., 2017; Arpit et al., 2017). We need a score that rewards the physics, not the fit. Metamorphic testing provides the machinery: it validates a program by checking that its outputs transform correctly under input transformations that the specification implies, without needing labels (Chen et al., 1998; Segura et al., 2016). For a physical predictor, the specification is the physics, and each metamorphic relation is one law that any genuinely causal estimator must satisfy.

We build a verifier that turns this into a certificate. The verifier trains a grid of models at increasing size and data budget, scores each trained model with a battery of metamorphic physics-relation tests to produce a Causal Robustness Score (CRS), and fits power laws of the score errors against compute to emit a verdict: causal scaling, stochastic parrot, or plateau. Two properties make the certificate trustworthy. First, we validate the battery against an oracle: a perfect constant-acceleration extrapolator, which is the exact solution of the training task, must score at least 0.95 on reflection equivariance and 0.90 on time dilation, so a low score is evidence of a causal defect rather than a test

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artifact. Second, we audit the detector itself, because a certificate is only as sound as the analysis that produces it.

Our experiments establish two load-bearing results on the ball benchmark. The battery rewards the true causal model, with the oracle scoring at least 0.95 and 0.90 on reflection equivariance and time dilation. On a GPU-scale grid of sixteen size-by-budget cells, the verifier certifies causal scaling: causal error falls with slope -0.111 ($R^2 = 0.734$) while task accuracy saturates at 0.92 with a detected plateau, the opposite of the stochastic-parrot signature. Reaching CRS 0.95 extrapolates to roughly 5.6 orders of magnitude more compute. The same data expose the audit's value: a constant-error bug in the detector produced a spurious parrot verdict, and the corrected detector reverses it. We report the verifier as the contribution and the audit as the condition that makes it trustworthy.

2 Problem formulation

2.1 Environments and training task

We consider two deterministic physics environments rendered to 48×48 grayscale frames. The ball environment simulates a bouncing ball in a box with gravity g, and the pendulum environment simulates a planar pendulum; both are parameterized per trajectory by $g \in [0.5, 2.0]$ and an initial speed range. Let $s_t \in \mathbb{R}^2$ be the state at frame t (the ball center, or the pendulum bob), and let $v_t = s_{t+1} - s_t$ be the per-frame displacement.

The training task is next-frame displacement prediction. Given a window of $k = 4$ rendered frames ending at time t, a model f_ϕ predicts $\hat{v}_t = f_\phi(\text{frames}_{t-3:t})$, trained with a smooth L1 loss against the true displacement. The task accuracy is $A = 1 - \text{mean}_t \, \|v_t - \hat{v}_t\| / \|v_t\|$ on a held-out validation set, so a memorizing model can push A high without learning the underlying dynamics.

2.2 Compute axis and grid

Each training run is a cell identified by a model width w and a data budget b. The model is a convolutional velocity predictor with two convolutions of width w followed by a two-layer MLP head of hidden width 64, so its parameter count $P(w)$ grows from 74,802 at $w = 8$ to 629,314 at $w = 64$. Optimizer steps follow the sublinear rule $S(b) = \lfloor 1400 \, (b/1500)^{0.45} \rfloor$ capped at 12,000, so the samples seen are $N = S(b) \cdot B$ with batch $B = 1024$. We define compute as $C = P(w) \cdot N$ and work on the log scale $x = \log_{10} C$. The grid covers $w \in \{8, 16, 32, 64\}$ and $b \in \{4000, 16000, 64000, 256000\}$, sixteen cells spanning $x \in [11.22, 12.89]$, roughly 1.7 orders of magnitude of compute.

3 Method

3.1 The metamorphic battery

Each test in the battery encodes one physics relation that any genuinely causal velocity estimator must satisfy, and scores the model by how covariantly its outputs transform under the corresponding input transformation. Every score lies in $[0, 1]$.

Reflection equivariance. Flipping an input frame vertically maps the true velocity (v_x, v_y) to $(v_x, -v_y)$, and flipping horizontally maps it to $(-v_x, v_y)$. The test flips each window, compares the predicted velocity with the flipped prediction, and scores with the mean relative error against a tolerance of 0.75, weighting the vertical flip twice the horizontal.

Time dilation. Resampling an input window at twice the frame rate must turn a predicted displacement $\hat{v}$ into $2\hat{v} + (0, g)$ on windows with no bounce, because the second difference of position vanishes under constant acceleration. The test evaluates the relation on bounce-free windows only and scores the mean relative error of the relation against a tolerance of 0.75.

Energy drift. Along a long trajectory, the total energy $E = \frac{1}{2}\|v\|^2 + V(s)$, with potential $V(s) = g(H - s_y)$ for a box of height H, must not drift beyond the simulator's own baseline. The test scores the excess drift of model-predicted velocities against a tolerance of 0.4.

Out-of-distribution extrapolation. The test evaluates relative velocity error on gravity values $g \in \{0.25, 0.4, 3.5, 6.0\}$ outside the training range, scoring against a tolerance of 0.75. A memorizing model fails here because the training distribution never contains these dynamics.

Closed-loop rollout. The model rolls out a trajectory open-loop for 30 steps from a real window, and the test scores the divergence of the rollout endpoint from the true trajectory, normalized by the box diagonal, against a tolerance of 0.8. Errors compound here, so this is the strictest test.

The Causal Robustness Score is the weighted sum $\text{CRS} = 0.25\,T_{\text{td}} + 0.15\,T_{\text{refl}} + 0.20\,T_{\text{energy}} + 0.20\,T_{\text{ood}} + 0.20\,T_{\text{roll}}$ of the five component scores.

Metamorphic test	Physics relation	Oracle score (enforced lower bound)
Reflection equivariance	$(v_x, v_y) \mapsto (v_x, -v_y)$ under flip	≥ 0.95
Time dilation	$\hat{v} \mapsto 2\hat{v} + (0, g)$ at $2\times$ rate	≥ 0.90
Energy drift	no excess $\frac{1}{2}\|v\|^2 + V(s)$ drift	exact solution
OOD extrapolation	correct velocity for unseen g	exact solution
Closed-loop rollout	30-step rollout tracks truth	exact solution

Table 1: The oracle validation. The constant-acceleration extrapolator is the exact solution of the displacement task on bounce-free windows, so its task accuracy is one by construction; the harness additionally enforces high scores on the two relations that are unit-testable against the oracle. A battery that failed this check would make every downstream certificate vacuous.

3.2 Validating the battery with an oracle

A battery that penalizes the true causal model would make every certificate vacuous, so the harness enforces a sanity condition: the constant-acceleration extrapolator $\hat{v}_t = 2v_{t-1} - v_{t-2}$, which is the exact solution of the displacement task on bounce-free windows, must score at least 0.95 on reflection equivariance and 0.90 on time dilation at render resolutions up to 32×32. This pins the battery to the physics: a model that is perfect on the training task is rewarded, and a low score must come from a causal defect, not a test artifact.

3.3 The scaling detector

For each cell we form the error metrics $e_{\text{crs}} = 1 - \text{CRS}$ and $e_{\text{acc}} = 1 - A$, both in $[10^{-3}, 1]$ so they live on a log-log axis. The detector fits $\log_{10} e$ against $x = \log_{10} C$ with a line (the power law $e = 10^{\alpha x + \beta}$) and a quadratic, and reports the slope α, the curvature γ, and both R^2 values. The verdict rules are explicit. Causal scaling requires $\alpha_{\text{crs}} < -0.1$ with no detected plateau. A plateau is detected when the quadratic curvature is positive and the quadratic fit improves on the linear fit by more than 0.02 in R^2, or the linear fit is weak. The stochastic-parrot signature fires when the accuracy error keeps falling with slope below -0.1 while the CRS curve flattens. Finally, the horizon extrapolates the CRS power law to the compute x^* at which e_{crs} reaches $1 - \tau$ for the target $\tau = 0.95$.

4 Experiments

4.1 Setup

All training runs use Adam with learning rate 3×10^{-3}, weight decay 10^{-4}, batch 1024, and at most $12{,}000$ steps. The battery uses the tolerances and weights above. The sixteen-cell grid ran as a single Kaggle kernel on a Tesla P100 GPU; training the sixteen ball cells took about 2.3 hours. The metamorphic battery is cheap to run: its evaluation, including the 30-step closed-loop rollouts and the out-of-distribution probes, added about 2 minutes across the ball cells and 5 minutes across the pendulum cells, roughly 1.5% of training time. The kernel is a private, self-contained Kaggle kernel whose handle we omit for anonymous review. The analysis is regenerable from committed logs by `papersrc/regen_kaggle_analysis.py`, and every number in this paper traces to that script's output.

4.2 Result 1: the battery rewards the true causal model

The validation harness enforces oracle scores of at least 0.95 on reflection equivariance and 0.90 on time dilation, and the oracle is exact on the training task by construction (Table 1). The contrast this buys is the point of the certificate: task accuracy alone cannot distinguish the oracle from a memorizer, because both can reach high A, but the battery separates them on physics the training loss never names.

4.3 Result 2: causal scaling is certified on the ball benchmark

Figure 1 shows all sixteen cells. Across $x \in [11.22, 12.89]$ the CRS rises from 0.667 to 0.786 while task accuracy rises from 0.864 to 0.922, and the two curves diverge at the top of the range: the accuracy error falls with slope -0.159 ($R^2 = 0.803$) but flattens, with positive quadratic curvature $+0.055$ and a detected plateau, while the CRS error keeps falling with slope -0.111 ($R^2 = 0.734$) and no detected plateau. The detector therefore emits the causal-scaling verdict. Task performance saturates before causal robustness does, which is the opposite of the stochastic-parrot signature: here the added compute is buying physics, not memorization. The horizon extrapolation places CRS 0.95 at $x^* \approx 18.46$, roughly 5.6 orders of magnitude ($\approx 3.7 \times 10^5\times$) beyond the measured range, under

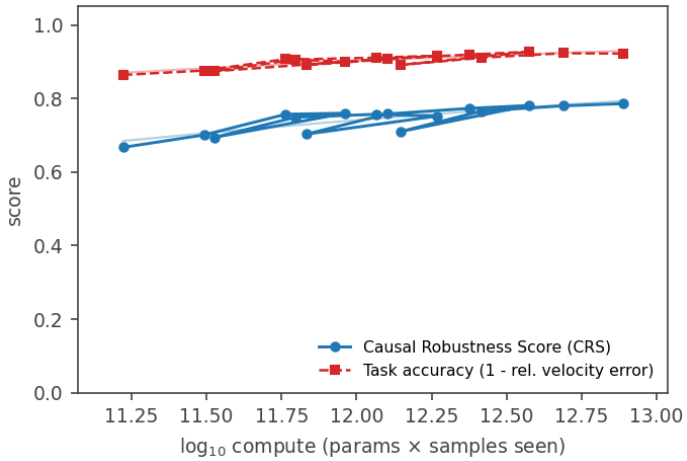


Figure 1: Causal scaling on the ball benchmark, sixteen cells ($n = 16$, one seed per cell, no cross-validation: each cell is an independent training run). The CRS (circles) climbs with compute while task accuracy (squares) saturates; thin curves are the fitted power laws and the dotted line marks the extrapolated CRS 0.95 horizon. The baseline is the fitted power law itself, and the divergence of the two curves at the top of the range is the load-bearing finding.

the assumption that the fitted power law continues; we report it as a theoretical upper bound on the compute needed to certify CRS 0.95, not as a prediction.

5 Discussion and limitations

The certificate is only as sound as the detector, and we found that out by auditing it. The CRS error originally fed the constant $1 - \tau = 0.05$ into the power-law fit instead of each cell's measured $1 - \mathrm{CRS}$, which made the CRS curve flat by construction and turned the same sixteen cells into a spurious stochastic-parrot verdict. The corrected detector, which fits the measured scores, reverses that verdict to causal scaling with slope -0.111 (Figure 2 in the appendix). The audit produced two things: a detector-level regression suite in the harness, and the rule that every number a certificate prints must trace to a measured quantity.

The limitations are concrete. The benchmark contains two toy environments, one architecture, and one seed per cell; a second independent run shows the low-compute cells are the noisiest (the smallest-cell CRS is 0.38 versus 0.67 across runs), so the low end of the grid should be read cautiously. The pendulum environment completes all sixteen cells in the full kernel run with accuracy 0.89 to 0.96, but the battery emits no pendulum certificate: the time-dilation relation assumes constant acceleration, which pendulum dynamics never satisfy (zero of $2{,}400$ sampled windows pass its filter). The defect that turned this into an undefined CRS, a mean over an empty selection, is fixed so the test scores zero rather than NaN, and instantiating a pendulum relation set is open work. The certificate is also borderline on the primary run: the measured CRS slope of -0.111 sits 0.011 from the detector's -0.1 decision boundary, and the CRS curve's quadratic improvement (0.019 in R^2) sits 0.001 from the 0.02 plateau bar, so stricter or looser decision thresholds flip the verdict (Appendix C). The CRS weights are not load-bearing (Appendix C). Sixteen cells span 1.7 orders of magnitude of compute, so the 5.6-order horizon is a theoretical upper bound that assumes the power law persists far beyond the measured range. The CRS is a benchmark certificate, not a deployment certificate: it certifies that the physics relations hold on the tested population, and an operator who changes the environment changes the certificate. The framework itself is environment-agnostic: each test is an input transformation paired with a physics relation and the verdict rules are unchanged across environments, so the certificate transfers to any environment with a simulator byre-instantiating the relations. Training takes about 2.3 hours per environment on a P100, and the verification overhead itself is small (1.5% of training time).

The forward direction is to make the certificate load-bearing for compute allocation: a detected accuracy plateau with a still-falling CRS says the next training run is worth it, and a certified CRS plateau says it is not. Calibrating the verdicts against causal interventions and ablating individual physics relations is the natural next check, because an auditor who trusts a certificate should be able to break it on purpose and watch it respond.

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A Detector-integrity ablation

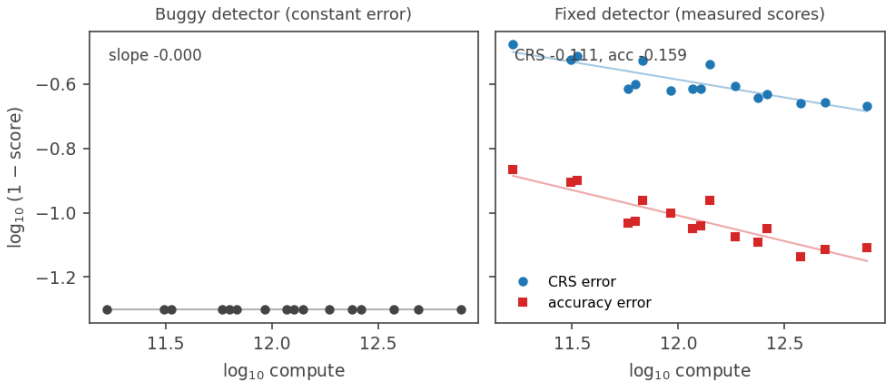


Figure 2: The same sixteen cells through two detectors. Left: the buggy detector fitted the constant error $1 - \tau = 0.05$, so its fitted CRS curve is flat by construction and the verdict was a spurious stochastic parrot. Right: the fixed detector fits each cell's measured $1 - \mathrm{CRS}$ and $1 - A$; the CRS error declines with slope -0.111 and the verdict reverses to causal scaling. This figure is the audit: a certificate that cannot distinguish a constant from a declining curve is vacuous, and the regression suite added by the audit enforces the distinction.

B Full cell table

width	params	steps	samples seen	$\log_{10} C$	CRS	accuracy
8	74,802	2,176	2,228,224	11.22	0.667	0.864
8	74,802	4,062	4,159,488	11.49	0.701	0.876
8	74,802	7,579	7,760,896	11.76	0.757	0.907
8	74,802	12,000	12,288,000	11.96	0.760	0.900
16	150,562	2,176	2,228,224	11.53	0.694	0.874
16	150,562	4,062	4,159,488	11.80	0.749	0.906
16	150,562	7,579	7,760,896	12.07	0.757	0.911
16	150,562	12,000	12,288,000	12.27	0.752	0.916
32	305,538	2,176	2,228,224	11.83	0.703	0.891
32	305,538	4,062	4,159,488	12.10	0.758	0.909
32	305,538	7,579	7,760,896	12.37	0.773	0.919
32	305,538	12,000	12,288,000	12.57	0.781	0.927
64	629,314	2,176	2,228,224	12.15	0.710	0.891
64	629,314	4,062	4,159,488	12.42	0.766	0.911
64	629,314	7,579	7,760,896	12.69	0.780	0.923
64	629,314	12,000	12,288,000	12.89	0.786	0.922

Table 2: All sixteen ball cells used in the analysis. Params come from the architecture, steps from the sublinear budget rule, samples seen is steps times the batch of 1024, and compute is the product of params and samples seen.

C Sensitivity analysis

We perturbed the two places the detector makes choices: the verdict thresholds and the CRS weights. Both sweeps use measured data only, the committed cells for the thresholds and the stored per-component scores of the second kernel run for the weights, and both are regenerable with `papersrc/sensitivity.py`.

Threshold sweep. We varied the slope bar (0.08, 0.10, 0.12), the plateau R^2 bar (0.01, 0.02, 0.03), the minimum R^2 (0.4, 0.5, 0.6), and the target CRS (0.90, 0.95, 0.98), 81 combinations. The causal-scaling verdict survives exactly the combinations with the slope bar at 0.08 or 0.10 and the plateau bar at 0.02 or 0.03 (36 of 81); it flips to the stochastic-parrot signature when the slope bar is tightened to 0.12 or the plateau bar is loosened to 0.01. The minimum-R^2 and target-CRS settings never change the verdict. The flips are close calls: the measured CRS slope of -0.111 is 0.011 from the -0.1 bar, and the CRS quadratic improvement of 0.019 in R^2 is 0.001 from the 0.02 bar.

Weight sweep. On the second run's stored component scores (same grid, independent seed), we renormalized the CRS weights with each weight scaled by 0.75 and 1.25 one at a time, plus 100 draws with each weight uniform in $[0.5, 1.5]$ times nominal. Per-cell CRS moves by at most 0.016, the fitted slope stays within $[-0.203, -0.193]$, and no perturbation changes the verdict. The weights are therefore not what the verdict depends on; the verdict is driven by the measured scores.

D Reproducibility

The kaggle preset (config `configs/kaggle.yaml`, run `python run.py -preset kaggle -device cuda`) produces the grid; a private, self-contained Kaggle kernel runs it on Kaggle (handle omitted for anonymous review and disclosed in the camera-ready). The analysis in this paper is regenerated from the committed kernel log `_logs.json` by `papersrc/regen_kaggle_analysis.py`, which uses the pipeline's own code paths, and the figures by `papersrc/make_figures.py`. The validation harness is `tests/test_core.py`; all sixteen tests pass, including the oracle bounds and the detector-level regressions added by the audit. The kernel self-heals across Kaggle GPUs: on a pre-Volta device it reinstalls a compatible PyTorch before training.